Sample Paper 05

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

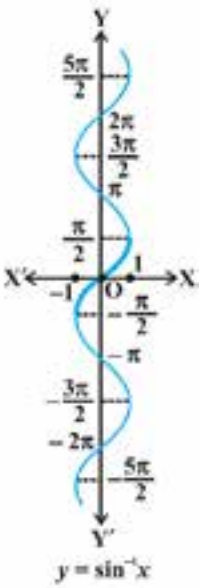
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the range of the graph of $y = \sin^{-1}x$ as shown in the image?



(a) [-1,1] (b) $[-\pi/2, \pi/2]$

(c) $[0, \pi]$ (d) $(-\infty, \infty)$

2. If $dy/dx = e^{-2y}$ and $y = 0$, when $x = 5$, then the value of x, when $y = 3$ is

(a) e^5 (b) $e^6 + 1$

(c) $\frac{e^6 + 9}{2}$ (d) $\log_e 6$

3. A constant force $\vec{F} = 2\vec{i} - 3\vec{j} + 2\vec{k}$ is acting on a particle such that the particle is displaced from the point $A(1, 2, 3)$ to the point $B(3, 4, 5)$. The work done by the force is

(a) 2 (b) 3

(c) $\sqrt{17}$ (d) $2\sqrt{51}$

4. The points (5, 2, 4), $(6, -1, 2)$ and $(8, -7, k)$ are collinear, if k is equal to

(a) -2 (b) 2

(c) 3 (d) -1

5. If x-coordinate of a point P of line joining the points and $R(5, 2, -2)$ is 4, then the z-coordinate of P is

(a) -2 (b) -1

(c) 1 (d) 2

6. Minimum value of the function $f(x) = x^2 + x + 1$ is

(a) 1 (b) 3

(c) $\frac{3}{4}$ (d) 4

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7. If $P(A)=\frac{1}{12}$, $P(B)=\frac{5}{12}$ and $P\left(\frac{B}{A}\right)=\frac{1}{15}$, then $P(A \cup B)$ is equal to

(a) $\frac{89}{180}$ (b) $\frac{90}{180}$

(c) $\frac{91}{180}$ (d) $\frac{92}{180}$

8. Range of the function $f(x)=\frac{x}{1+x^2}$ is

(a) $(-\infty,\infty)$ (b) $[-1,1]$

(c) $\left[-\frac{1}{2},\frac{1}{2}\right]$ (d) $[-\sqrt{2},\sqrt{2}]$

9. If $A=\{1,2,3\}$ and $B=\{2,3,4\}$, then which of the following relations is a function from A to B?

(a) $\{(1,2),(2,3),(3,4),(2,2)\}$

(b) $\{(1,2),(2,3),(1,3)\}$

(c) $\{(1,3),(2,3),(3,3)\}$

(d) $\{(1,1),(2,3),(3,4)\}$

10. The order of the differential equation $\left[1+\left(\frac{dy}{dx}\right)\right]^{3/2}=\frac{d^2y}{dx^2}$ is

(a) 1 (b) 2

(c) 3 (d) 4

11. The function $f(x)=2+4x^2+6x^4+8x^6$ has

(a) only one maxima

(b) only one minima

(c) no maxima and minima

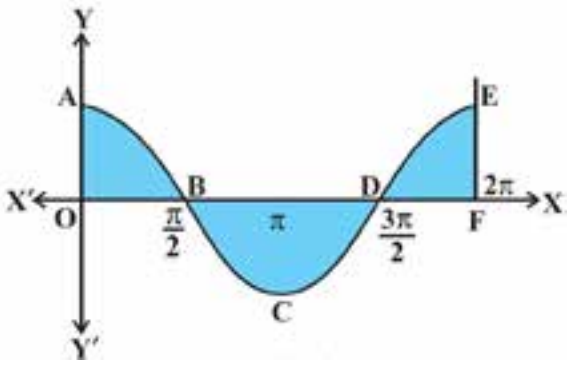
(d) many maxima and minima

12. $2\sin^{-1}x=\sin^{-1}(2x\sqrt{1-x^2})$ holds good for all

(a) $|x|\le 1$ (b) $1\ge x\ge 0$

(c) $|x|\le\frac{1}{\sqrt{2}}$ (d) none of these

13. In Fig., the shaded regions are bounded by which of the following?



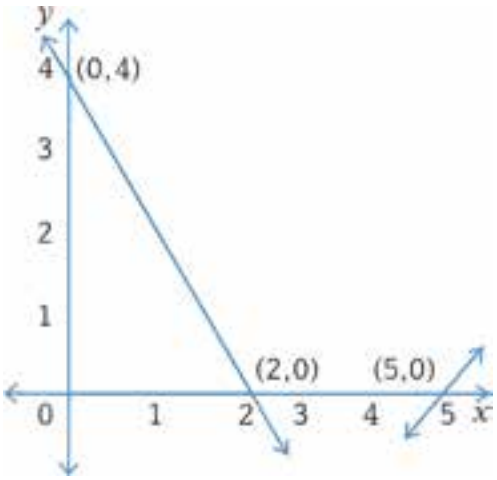
(a) The curve passing through points A,B,C,D,E and the X-axis between x=0 and x=2π

(b) Only the straight line y=0 and the X-axis

(c) The vertical lines x=-π and x=π

(d) A semicircle between 0 and 2π

14. For the same LPP with objective function $Z=3x+2y$, what is the minimum value of Z and at which point does it occur?



(a) Z=8 at (0,4) (b) Z=15 at (5,0)

(c) Z=6 at (2,0) (d) No minimum exists

15. The existence of the unique solution of the system of equations $x+y+z=\beta$; $5x-y+\alpha z=10$ and $2x+3y=6$ depends on

(a) α only (b) β only

(c) Both α and β (d) Neither β nor α

16. At $x=\frac{3}{2}$, the function $f(x)=\frac{|2x-3|}{2x-3}$ is

(a) continuous (b) discontinuous

(c) differentiable (d) non-zero

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17. If $f(x)=\begin{cases}\frac{\log x}{x-1}, & \text{if } x\neq 1\\ k, & \text{if } x=1\end{cases}$ is continuous at $x=1$, then the value of k is

(a) 0 (b) -1

(c) 1 (d) e

18. Given two vectors

$\vec{a}=2\hat{i}-3\hat{j}+6\hat{k}$, $\vec{b}=-2\hat{i}+2\hat{j}-\hat{k}$

and $\lambda=\frac{\text{the projection of } \vec{a} \text{ on } \vec{b}}{\text{the projectin of } \vec{b} \text{ on } \vec{a}}$

then the value of λ is

(a) $\frac{3}{7}$ (b) $\frac{7}{3}$

(c) 3 (d) 7

19. **Assertion :** $I=\int_0^{\frac{\pi}{2}}\sqrt{\tan x}\,dx=\frac{\pi}{\sqrt{2}}$

Reason : $\tan x=t^2$ makes the integrand in I as a rational function.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

20. **Assertion:** $f(\theta)=\begin{vmatrix}\cos(\theta+\alpha) & \cos(\theta+\beta) & \cos(\theta+\gamma)\\ \sin(\theta+\alpha) & \sin(\theta+\beta) & \sin(\theta+\gamma)\\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta)\end{vmatrix}$ is independent of θ

Reason: If $f(\theta)=c$ then $f(\theta)$ is independent of θ

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Write the direction ratios of the vector $3\vec{a}+2\vec{b}$, where $\vec{a}=\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=2\hat{i}-4\hat{j}+5\hat{k}$.

OR

Find the unit vector in the direction of the sum of vectors $2\hat{i}+3\hat{j}-\hat{k}$ and $4\hat{i}-3\hat{j}+2\hat{k}$.

22. Let R is the equivalence relation in the set $A=\{0,1,2,3,4,5\}$ given by $R=\{(a,b):2 \text{ divides } (a-b)\}$. Write the equivalence class $[0]$.

23. Find the vector equation of the line which passes through the point (3,4,5) and is parallel to the vector $2\hat{i}+2\hat{j}-3\hat{k}$.

24. Prove that if E and F are independent events, then the events E and F' are also independent.

25. Find $\int\frac{\sin^6 x}{\cos^8 x}dx$.

OR

Evaluate $\int\frac{dx}{\sin^2 x\cos^2 x}$.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. Find the value of k, so that the following functions is continuous at $x=2$.

$f(x)=\begin{cases}\frac{x^3-x^2-16x+20}{(x-2)^2}, & x\neq 2\\ k, & x=2\end{cases}$

OR

Find $\frac{dy}{dx}$ at $x=1$, $y=\frac{\pi}{4}$ if $\sin^2 y+\cos xy=K$.

27. The volume of a sphere is increasing at the rate of 8 cm^3/s. Find the rate of which its surface area is increasing when the radius of the sphere is 12 cm.

OR

Show that the function $f(x)=x^3-3x^2+6x-100$ is increasing on R.

28. Write a unit vector perpendicular to both the vectors $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$.

29. Write the value of $\cos^{-1}\left(\frac{1}{2}\right)-2\sin^{-1}\left(-\frac{1}{2}\right)$.

30. Find the maximum value of $\begin{vmatrix}1 & 1 & 1\\ 1 & 1+\sin\theta & 1\\ 1 & 1 & 1+\cos\theta\end{vmatrix}$

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31. Evaluate $\int \frac{2\cos x}{3\sin^2 x}\,dx$.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Show that the differential equation $\left[x\sin^2\left(\frac{y}{x}\right) - y\right]dx + x\,dy = 0$ is homogeneous.
Find the particular solution of this differential equation, given that $y = \frac{\pi}{4}$, when $x = 1$.

33. Maximise $Z = 5x + 3y$ subject to the constraints: $3x + 5y \le 15$; $5x + 2y \le 10$, $x \ge 0$, $y \ge 0$.

OR

Minimize $Z = 3x + 5y$ such that $x + 3y \ge 3$, $x + y \ge 2$, x, $y \ge 0$.

34. Show that the line lines

$$\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(3\hat{i} - \hat{j}) \text{ and}$$

$$\vec{r} = (4\hat{i} - \hat{k}) + \mu(2\hat{i} + 3\hat{k})$$

intersect. Also, find their point of intersection.

35. Find the value of $\int \frac{\cos x}{(1+\sin x)(2+\sin x)}\,dx$.

OR

Prove that $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$. and hence evaluate $\int_0^{\pi/2} \frac{x}{\sin x + \cos x}\,dx$.

SECTION - E

Case study based questions are compulsory.

36. The U.S. Constitution directs the government to conduct a census of the population every 10 years. Population totals are used to allocate congressional seats, electoral votes, and funding for many government programs. The U.S. Census Bureau also compiles information related to income and poverty, living arrangements for children, and marital status. The following joint probability table lists the probabilities corresponding to marital status and sex of persons 18 years and over.14

Sex	**Marital Status**			
	(R)	(N)	(W)	(D)
(M)	0.282	0.147	0.013	0.043
(F)	0.284	0.121	0.050	0.060

$R \Rightarrow$ Married
$N \Rightarrow$ Never married
$W \Rightarrow$ Widowed
$D \Rightarrow$ Discovered or separated
$M \Rightarrow$ Male
$F \Rightarrow$ Male

Suppose a U.S. resident 18 years or older is selected at random.

(i) Find the probability that the person is female and widowed.

(ii) Suppose the person is male. What is the probability that he was never married?

(iii) Suppose the person is married. What is the probability that the person is female?

37. A market analysis is a quantitative and qualitative assessment of a market. It looks into the size of the market both in volume and in value, the various customer segments and buying patterns, the competition, and the economic environment in terms of barriers to entry and regulation.



Based on the past marketing trends and his own experience, marketing expert suggested to the

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concerned the segments of market for their products as follows:

The first segment consisted of lower income class, the second segment that of middle income and the third segment that of high income. The data based on the income of the consumers was readily available. During a particular month in particular year, the agent reported that for three products of the company the following were the sales: There were 200 customers who bought all the three products, 240 customers who bought I and III, 60 customers only products II and II and 80 customers only products only III regardless of the market segmentation groups.

Based on the market segmentation analysis, for product I, the percentage for the income groups are given as (40%, 20% and 40%), for product II (30%, 20% and 50%), for product III (10%, 50% and 40%).

(i) Taking the suitable variable form the system of equation that represent given problem.

(ii) Using matrix method, find out the number of persons in the lower income, middle income and higher income class in the region referred.

38. In apparels industries retailers have an interesting conundrum facing them. On one hand, consumers are more drawn to hot promotional deals than ever before. The result of this is that they sell more units (of product) for less money, and this adversely impacts comp store sales.



Arvind Fashions knows that the it can sell 1000 shirts when the price is ₹ 400 per shirt and it can sell 1500 shirts when the price is ₹ 200 a shirt. Determine

(i) The price function

(ii) The revenue function

(iii) The marginal revenue function.

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